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PALEOMAGNETIC INVESTIGATION OF THE CAMBRIAN WAYNESBORO AND ROME FORMATIONS OF THE VALLEY AND RIDGE PROVINCE OF THE APPALACHIAN MOUNTAINS

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Abstract. The Lower to lower Middle Cambrian Waynesboro and Rome formations yield paleomagnetic directions with east-southeast to southeast declinations and shallow positive inclinations; the corresponding pole positions fall at 28°N, 165°E (dp = 11°, dm = 22°) and 39°N, 142°E (dp = 6°, dm = 12°), respectively. The sampling concentrated on red sandstones and siltstones and crossed several styles of deformation and changes in structural trend in the central Appalachians. Thermal and alternating field demagnetizations were analyzed with the aid of vector diagrams and the combined results of the two formations yield a positive fold test at the 99% confidence level. Since the mean directions from these formations deviate substantially in a clockwise sense from directions predicted by previously published paleomagnetic poles obtained from Cambrian rocks, the possibility of local rotations of the sampling sites has been considered. Two arguments, however, can be raised against such rotations. First, there is no geological evidence for rotations of the Tennessee thrust sheets relative to one another or relative to the craton. Second, there are directions from Ordovician and late Precambrian-Early Cambrian formations in the Tennessee thrust sheets that do not deviate in a clockwise sense from those predicted by results from the craton. Remagnetization and nondipole behavior of the Cambrian geomagnetic field have been suggested as alternative explanations for the North American Lower and Middle Cambrian paleomagnetic poles but are considered unlikely causes for some of the new results from this and a companion paper. Instead, Cambrian apparent polar wander is proposed to explain the distribution of the pole positions.

Introduction

The documentation of the Cambrian segment of the North American apparent polar wander path (APWP) has been hampered by an apparent scatter of Cambrian paleomagnetic poles published in the last decade. As has been discussed by Van der Voo [1978], Cambrian poles from diverse structural and geographic provinces of North America fall in a broad zone between the equator at about 160°E and the pole obtained for the Abrigo Formation at 59°N, 89°E by Elston and Bressler [1977]. French [1976] found that a preliminary pole position for the Cambrian Rome Formation in Tennessee fell between the two extremes of the observed streak of about 60° of arc. However, this pole position was suspect because it was derived from rocks that had been transported by the movement along thrust faults characteristic of this part of the Valley and Ridge Province of the Appalachians. In his study, French [1976] could not preclude the possibility that rotations occurred during this movement thereby giving rise to anomalous pole positions. We have extended the sampling in Tennessee and also have collected the correlative Waynesboro Formation in southern Pennsylvania and Maryland (Figs. 1 & 2). The extended sampling, structural considerations of the thrust sheet geometry in Tennessee as well as some recent Ordovician paleomagnetic results from the same areas [Van der Voo and French, 1977; Watts and Van der Voo, 1979] allow us to address the problem of possible rotations and to report a pole for the Waynesboro and Rome formations we feel is representative of the North American craton.

Geology

Through the length of the Valley and Ridge Province of the Appalachian Mountains, from Pennsylvania to Georgia, a heterogeneous sequence of variegated (dominantly red) sandstone, siltstone, and shale interbedded with dolomite crops out in a series of well-defined strike belts. Known as the Waynesboro Formation north of Roanoke, Virginia, and the Rome Formation south of Roanoke [King, 1950], this dominantly terrigenous sequence is of Early to early Middle Cambrian age and ranges in thickness from 200 to 700 m. The Rome and Waynesboro formations are regarded as equivalent by King [1950], who suggests that at some future date the term 'Waynesboro' may be dropped and the term 'Rome' applied ubiquitously. Gathright and Nystrom [1974], however, point out that the Waynesboro Formation appears to be equivalent only to two clastic units in what is mapped as the lower Rome Formation in the Ashby Gap Quadrangle, Virginia. In eastern Tennessee the boundary between the Lower and Middle Cambrian is drawn between the Rome and the overlying Pumpkin Valley Formation [Rodgers, 1953]. This distinction appears to be possible only in Tennessee, since the Rome and Waynesboro formations are sparingly fossiliferous and poorly exposed elsewhere. Hence the possibility of age variation along strike cannot be discounted.

The Waynesboro and Rome formations represent an episode of terrigenous deposition between great thicknesses of carbonate sediments. Characterized by abundant mud cracks, ripple marks, laminar bedding, and a trilobite fauna, the Rome (and Waynesboro) Formation is interpreted to have been deposited in a tidal mudflat [Milici,

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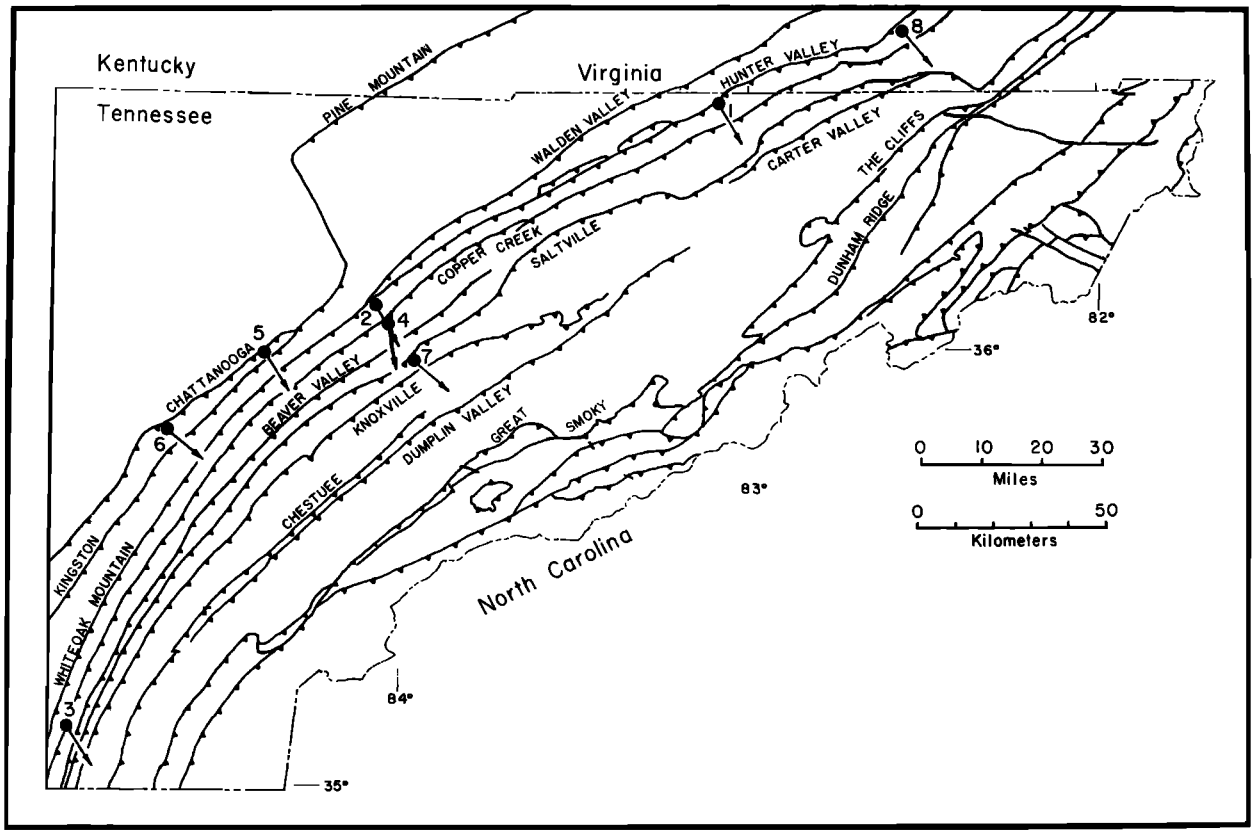


Fig. 1. Generalized geological map of eastern Tennessee showing major thrust faults and Rome sampling sites. For each sampling site the site mean paleomagnetic declinations are illustrated by the arrows.

1973], probably near the Cambrian margin of North America.

The Rome and Waynesboro formations are among the lowest stratigraphic units in the Valley and Ridge Province of the Appalachians and have therefore undergone considerable deformation. The style of deformation, however, varies greatly from Pennsylvania through Tennessee.

The structural geology of the Blue Ridge and Valley and Ridge provinces of southern Pennsylvania, Maryland, West Virginia, and Virginia has been described by Cloos [1950], Slaughter and Darling [1962], and Cloos [1971]. The Waynesboro outcrop pattern in southern Pennsylvania and Maryland (Figure 2) is dominated by the South Mountain anticlinorium which extends roughly from the western edge of Frederick Valley almost to Hagerstown, Maryland [Cloos, 1958]. The intensity of deformation decreases from east to west, having the greatest intensity in the Precambrian basement and Cambrian (?) Chilhowee Group [Cloos, 1971]. Although local variations may be considerable, Cloos [1971] has demonstrated the existence of a structural fabric that is consistent throughout the anticlinorium and that extends 200 m (320 k) from Pennsylvania to Virginia. The folding is asymmetric on all scales, with anticlines having gently dipping eastern flanks and steeply dipping to overturned western flanks. Fold axes are horizontal to gently plunging. The structural fabric is present from the Precambrian basement up through the Silurian and is manifested by the consistency of orientations of deformed oolites, lineations associated with the intersection of bedding and cleavage, and the growth of secondary minerals in fractures and veins. There is no doubt that all rocks in the South Mountain anticlinorium, including the Waynesboro Formation, have undergone a finite strain, though there are considerable variations in the intensity which depend on rock type and position in the anticlinorium.

The style of deformation of the Rome Formation in the Valley and Ridge Province of southern Virginia and eastern Tennessee is considerably different (Figure 1). The Paleozoic cover here is detached from the basement and deformed by a series of thrust faults [Cloos, 1971; Rodgers, 1953; Milici, 1973], and nowhere is the contact with basement observed. A principal surface of detachment formed within the Rome Formation. As a result, most outcrops of the Rome in this area occur on the leading edges of the great thrust sheets, and the Rome is the lowest widely exposed stratigraphic unit in this structural province. This role in the structural evolution of the area has often resulted in severe deformation of the Rome, although the amount of deformation varies considerably. Because of the association with the thrusting the base of the Rome is not exposed, and many of the sections may be repeated. These factors contribute great uncertainty to any estimate of the thickness of the Rome Formation, and they preclude any attempts at local correlations between individual outcrops.

In southern Virginia and eastern Tennessee

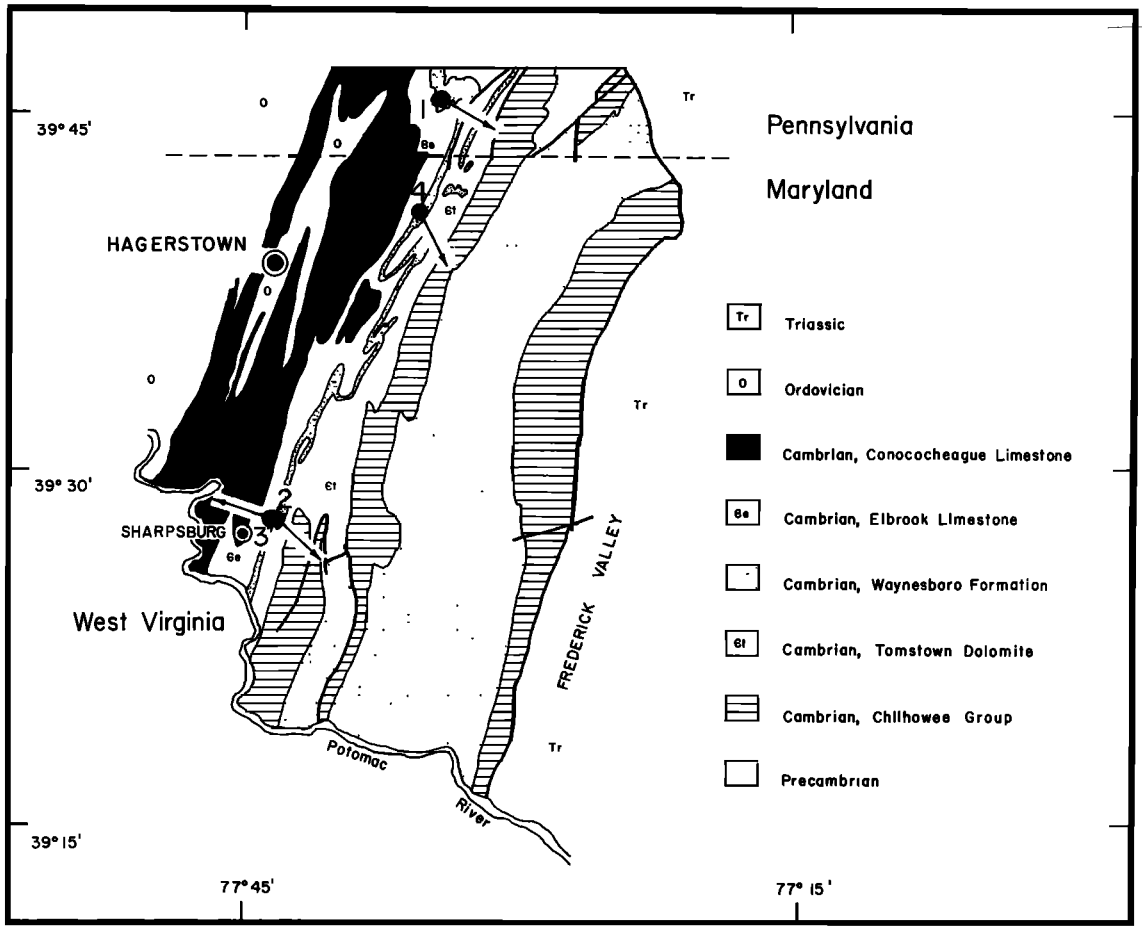


Fig. 2. Generalized geological map of the South Mountain anticlinorium in Maryland and southern Pennsylvania with Waynesboro sampling sites. The site mean declinations are illustrated by the arrows.

there is no apparent regional structural fabric such as can be demonstrated for the Valley and Ridge of southern Pennsylvania and Maryland. Cleavage is locally developed in some formations but cannot always be directly related to folding [Swingle, 1973]. Each of the thrust sheets has undergone internal folding, but the intensity and style varies considerably [Swingle, 1973; Rodgers, 1953]. Fold axes tend to parallel the strike of the great thrust faults, and the traces of thrust faults are remarkably parallel (Figure 1).

Sampling

Oriented samples from the sandstone and siltstone members of the Rome and Waynesboro formations were collected from the sites indicated in Figures 1 and 2. The sampling was concentrated in Maryland and Tennessee. Outcrops in West Virginia and Virginia, described by King [1950] Amato [1974], Gathright and Nystrom [1974], and McGuire [1976], were examined but were found to have undergone such intense deformation that bedding was obscured by cleavage and attitudes were uncertain at best. In many outcrops in West Virginia and Virginia the bedding in the terrigenous units was completely obscured by cleavage. Suitable (i.e., relatively unstrained) outcrops were found in Maryland, Pennsylvania, Tennessee, and southernmost Virginia. Every attempt was made to locate beds with opposing dips so a fold test could be carried out.

Twenty-five oriented samples were gathered from four sites in the Waynesboro Formation (Figure 2). Site 1 is near the type locality of the Waynesboro Formation [Stose, 1906], in Waynesboro, Pennsylvania. The beds at this site are nearly flat lying as the site is apparently situated at the hinge of a large fold (S. Root, personal communication, 1977). Site 2 is at the hinge of an anticline; the top of the section was evident from abundant ripple marks and mud cracks in the red siltstone. Site 3 is in the steeply dipping northwest limb of the same anticline. Site 4 is in nearly horizontal beds with a vertical cleavage.

Seventy-nine oriented samples were collected from eight sites in the Rome Formation in Tennessee (Figure 1). Site 1 is in a series of drag folds in the Rome where it is possible to apply a fold test directly. Although sites 2 through 8 in the Rome all have dips to the southeast, the dips varied between 40° and 78°, affording the possibility of an attitude test in the Rome Formation. Many sites have small internal variations in bedding attitude.

Magnetization of the Rome and Waynesboro Formations

Both oriented block samples and drill cores were collected. In the laboratory, one core was taken from each block sample. Cores were sliced into cylinders approximately 2.5 cm in diameter and 2.3 cm in length. The natural remanent magnetization (NRM) was measured with a Schonstedt SSM-1A spinner magnetometer and a ScT superconducting magnetometer capable of measuring samples with a magnetization of 10^{-5} A m^{-1} or more.

To isolate secondary and characteristic components of the NRM, the specimens were subjected to thermal, alternating field (AF), and chemical demagnetization. After each treatment and measurement the result was plotted on orthogonal projections [Zijerveld, 1967]. Treatment was carried out until it was evident that all components within the sample had been identified. The various components were resolved by examining the Zijderveld plots of linear demagnetization trajectories defined by three or more points and applying vector subtraction to the results that best defined linear trends. Some of the orthogonal projections are illustrated in Figure 3 for the Waynesboro Formation and Figure 4 for the Rome Formation.

Alternating field and thermal demagnetization were successful in isolating the characteristic components. Chemical leaching was not successful because of the impermeable nature of the samples investigated. Although the samples were slotted to facilitate acid penetration [Henry, 1979], it was found that after more than 1000 hours of leaching in 10 N HCl, over 60% of the NRM remained. Also, the magnetization removed during leaching did not follow linear trajectories in the orthogonal projections of the demagnetization results. For this reason, chemical leaching is regarded as an unsuitable technique for isolating the various magnetic components of the Rome and Waynesboro sandstones and siltstones.

Alternating field demagnetization was performed using a Schonstedt GSD-1 unit which has a maximum peak field of 100 mT. This treatment resulted in the removal of secondary components of low coercivity but did not succeed in breaking down the characteristic component (Figure 4a).

Thermal demagnetization was done in a Schonstedt TSD-1 unit or in a noninductive furnace in feedback-controlled helmholtz coils. Thermal demagnetization of the Waynesboro and Rome specimens reveals secondary and characteristic components (Figures 3a and 4b). Figures 4a and 4b show the result of AF and thermal demagnetization of two specimens from the same core. Comparison of these two figures shows removal of a low-coercivity or low blocking temperature component and illustrates that the magnetization remaining after treatment at 100 mT is parallel to the characteristic directions obtained from thermal demagnetization. The directions of magnetization remaining after 100 mT were used therefore in the calculation of the mean directions and pole positions. However, because of the better resolution of the magnetic components, thermal demagnetization was applied to a majority of the specimens. It was found that thermal demagnetization after AF demagnetization on a given specimen resulted in very

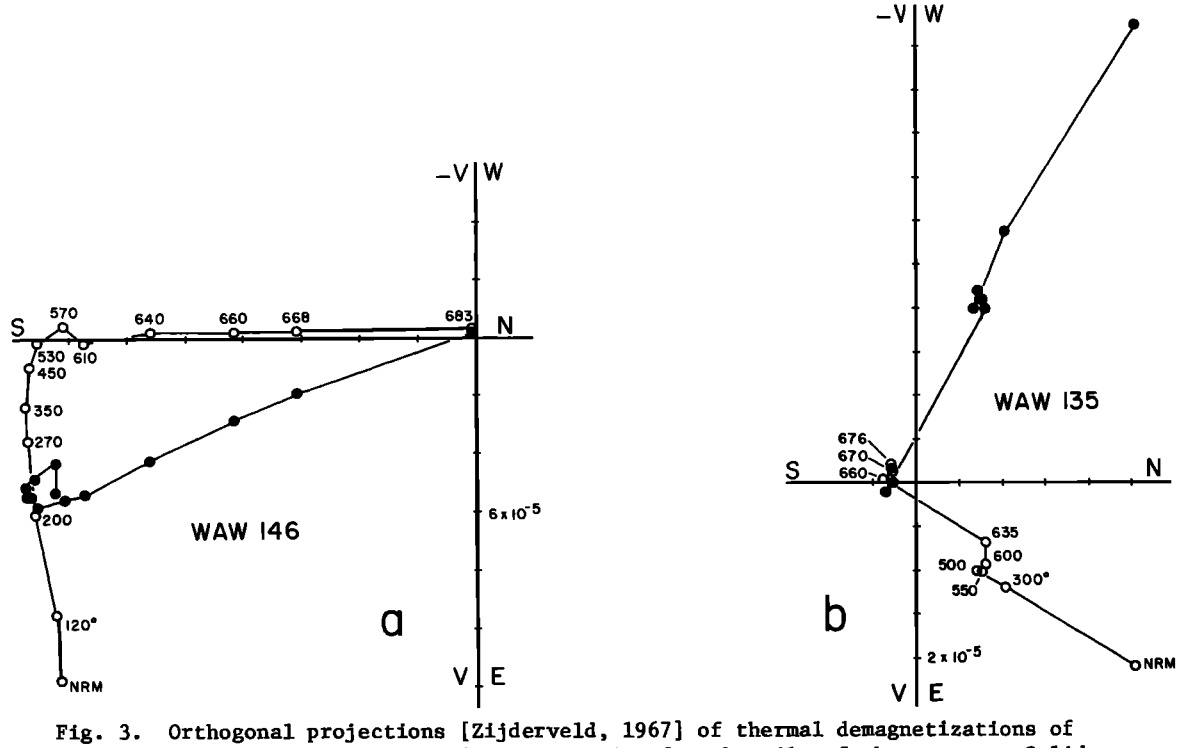


Fig. 3. Orthogonal projections [Zijderveld, 1967] of thermal demagnetizations of Waynesboro Formation specimens after correction for the tilt of the strata. Solid circles represent projections on the horizontal plane and open circles those on the north-south vertical plane. Units indicated for the axes are in electromagnetic units per cubic centimeter (10^{-3} emu/cm^3 = 1 A m^{-1}). (a) A low blocking temperature secondary component and a high blocking temperature component characteristic of the Waynesboro Formation and (b) a normal polarity component.

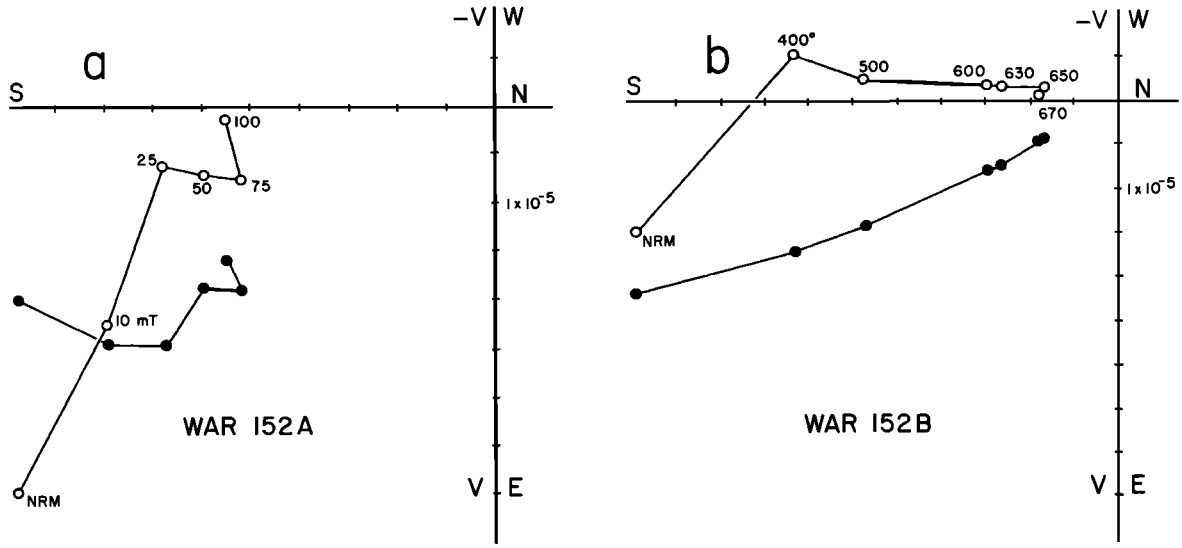


Fig. 4. Orthogonal demagnetization projections of two specimens from same core taken from the Rome Formation after correction for the tilt of the strata. Symbols and units are the same as those in Figure 3. (a) the Results of AF (alternating field demagnetization) with the level of treatment given in milliTeslas and (b) the result of thermal demagnetization.

noisy orthogonal plots and could not be interpreted.

Figures 5a and 5c are equal area projections of the characteristic directions and site mean directions before structural correction and Figures 5b and 5d are the projections of the characteristic and site mean directions after correction. The three normal polarity directions are from site 3 (Figure 2), the northwest dipping limb of an asymmetric anticline. The characteristic components are distinguished from secondary magnetizations by their persistence at high blocking temperatures (Figure 3b). Figure 3b shows that although the normal polarity component is revealed by a linear trajectory in the orthogonal projection, it does not decay toward the origin. The nature and direction of the magnetization remaining after 676°C could not be determined, but its existence suggests that perhaps the normal polarity components are overprints.

Comparing Figures 5a to 5b, the in situ characteristic directions are scattered (k = 4), but the clustering improves after structural correction is applied (k = 7). The site mean directions are listed in Table 1, along with the associated statistics, and are plotted in Figures 5c and 5d, before and after structural correction, respectively. Inverting the mean direction of site 3 so that its polarity becomes reversed as well, the site mean directions of the Waynesboro Formation cluster better after structural correction (k = 5 before and k = 19 after correction). However, this increase in precision parameter is not statistically significant at the 95% confidence level [McElhinny, 1964].

Figures 6a and 6b are equal area projections of the Rome characteristic directions, before and after structural correction. The site mean directions are given in Table 1 and are plotted in Figures 6c and 6d, before and after correction. The characteristic directions cluster better after structural correction (k = 10 before, k = 12 after), as do the site mean directions (k = 14 before, k = 22 after), but in neither case is the improvement significant at the 95% confidence level. The only clearcut within-site fold test (Rome site 1, Table 1), shows an increase in the precision parameter, but this increase is not significant at the 95% confidence level.

Although the scatter of the characteristic directions and the site mean directions of the Rome and Waynesboro formations is reduced after structural correction in all instances, the decrease in the scatter cannot be regarded as a positive fold test at the 95% level of confidence. This is probably due to the substantial number of outcrops within each formation of similar attitude. Three of the four Waynesboro outcrops are subhorizontal, and all of the Rome outcrops dip to the east. An overall fold test, involving pole positions for the site mean directions of the Waynesboro and Rome formations, is possible. Figure 7a shows the distribution of site mean poles of the Rome and Waynesboro formations before structural correction, and Figure 7b shows the distribution of site mean poles after structural correction. Statistically, the structural correction increases the K of the distribution of poles from K = 4 to K = 21 (Table 1). The improved clustering of the Waynesboro and Rome site mean poles is significant at the 99% level of confidence. This overall fold test indicates a predeformation (pre-Allegheny) time of acquisition of the characteristic directions of the Rome and Waynesboro formations; it does not, of course, prove a Cambrian age of magnetization.

The in situ low-coercitivity, low blocking temperature directions are grouped around the present-day field direction. These directions do not pass the fold test, and they are interpreted as secondary magnetizations of fairly recent origin.

Thin-section examination with transmitted

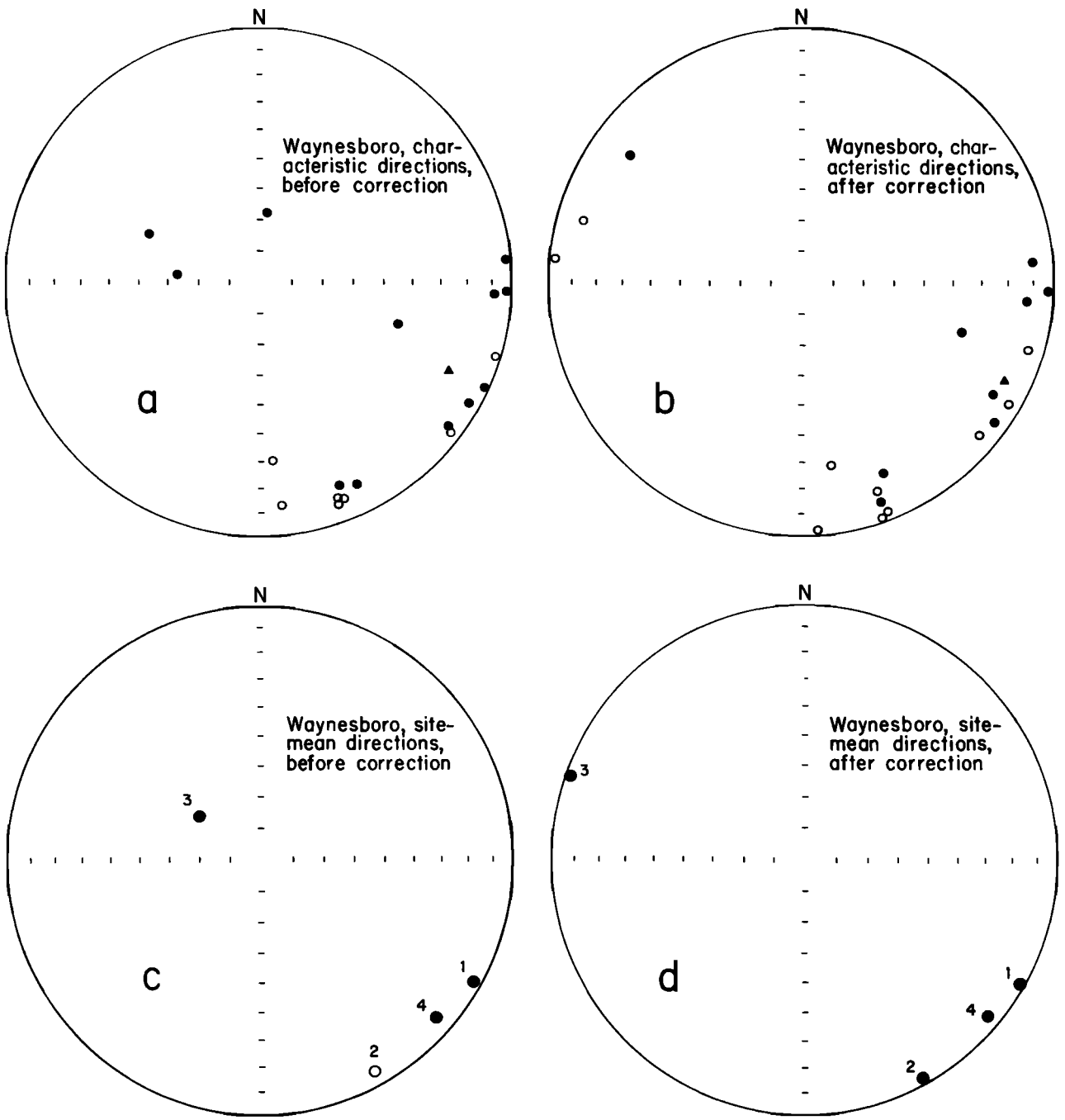


Fig. 5. Equal area projections of the Waynesboro Formation characteristic and site mean directions before (Figures 5a and 5c) correction and after (Figures 5b and 5d) correction for the tilt of the strata. Circles denote thermal demagnetization results and triangles denote AF demagnetization results. Open (solid) symbols represent projections on the upper (lower) hemisphere.

light of the Waynesboro and Rome sandstones and siltstones revealed that the history of the hematite growth is complex. In all sections some hematite (specularite) appears as irregular, detrital grains concentrated in discrete bedding planes. In some sections, however, hematite can also appear as a pigment on the detrital grains or as a filling of interstices (cement). As in most red bed sequences, at least part of the characteristic remanence is probably a chemical remanent magnetization acquired after deposition. A small percentage of heavy minerals (muscovite, biotite, hornblende) is present, and some of the hematite is associated with these minerals, indicating that at least a small percentage of the iron oxide is due to alteration of the heavy mineral fraction. In all slides investigated, the major portion of the hematite appears to be detrital. Although magnetite was not observed in any abundance, its presence is suggested by the low-coercivity components found in most of the specimens.

TABLE 1. Site Mean Directions, Statistics, and Pole Positions

Site	Location, deg	R_s	Before Tilt Correction			After Tilt Correction				Pole Position	dp	dm
			Decl	Incl	k	Decl	Incl	k	α_{95}			
	Waynesboro Formation: Characteristic Site Means											
Site 1	39°45.4'N, 77°34.7'W	6/7	120°	+3°	5	120°	+2°	5	32°	21°N, 171°E	16°	32°
Site 2	39°28.0'N, 77°43.4'W	4/5	152°	-6°	8	152°	0°	8	36°	43°N, 142°E	18°	36°
Site 3	39°28.0'N, 77°43.4'W	3/3	304°	+66°	14	289°	+2°	14	34°	16°N, 180°E	17°	34°
Site 4	39°41.3'N, 77°35.0'W	7/10	132°	+7°	7	131°	+6°	7	24°	28°N, 161°E	12°	24°
	Rome Formation: Characteristic Site Means											
Site 1	36°34.1'N, 83°05.2'W	9/9	142°	+51°	9	150°	+17°	11	16°	37°N, 135°E	8°	16°
Site 2	36°02.5'N, 84°06.2'W	39/39	154°	+42°	15	150°	-1°	14	6°	45°N, 140°E	3°	6°
Site 3	35°08.9'N, 84°57.5'W	6/6	181°	+50°	18	146°	+14°	29	12°	37°N, 140°E	6°	13°
Site 4	36°04.5'N, 84°02.7'W	4/5	185°	+44°	37	167°	+6°	18	22°	49°N, 116°E	11°	22°
Site 5	35°59.3'N, 84°25.1'W	2/2	223°	+78°	--	147°	+26°	--	--	31°N, 134°E	--	--
Site 6	35°59.8'N, 84°41.3'W	4/5	87°	+81°	15	130°	+18°	15	25°	25°N, 152°E	14°	26°
Site 7	35°59.8'N, 83°57.0'W	7/8	133°	+67°	16	132°	-12°	14	14°	37°N, 164°E	7°	14°
Site 8	36°47.3'N, 82°34.0'W	5/5	116°	+57°	11	142°	-4°	11	24°	41°N, 152°E	12°	24°

Mean	R_s	Before Tilt Correction			After Tilt Correction				Mean pole	dp	dm
		Decl	Incl	k	Decl	Incl	k	α_{95}			
Waynesboro, mean of all samples	20/25	132°	-7°	4	128°	+2°	7	14°	28°N, 165°E	7°	14°
Waynesboro, mean of all sites	4/4	133°	-14°	5	128°	+2°	19	22°	28°N, 165°E	11°	22°
Waynesboro, mean of the poles	4/4			8			20	21°	28°N, 165°E		
Rome, mean of all samples	76/79	153°	+52°	10	147°	+3°	12	5°	42°N, 142°E	3°	5°
Rome, mean of all sites	8/8	156°	+62°	14	146°	+8°	22	12°	39°N, 142°E	6°	12°
Rome, mean of the poles	8/8			6			37	9°	38°N, 142°E		
Waynesboro and Rome, mean of all poles	12/12			4			21	10°	36°N, 150°E		

Decl and Incl are declination and inclination, respectively, in degrees; R_s is the ratio of samples or sites used to derive a particular mean direction or pole to the total number of samples or sites analyzed; k is the ~~est~~imate of Fisher's precision parameter [McElhinny, 1973] for site mean, sample mean, or pole mean; and α_{95} is the associated radius of the cone of 95% confidence; dp and dm are the semiaxes of the oval of 95% confidence about the poles calculated from mean paleomagnetic directions.

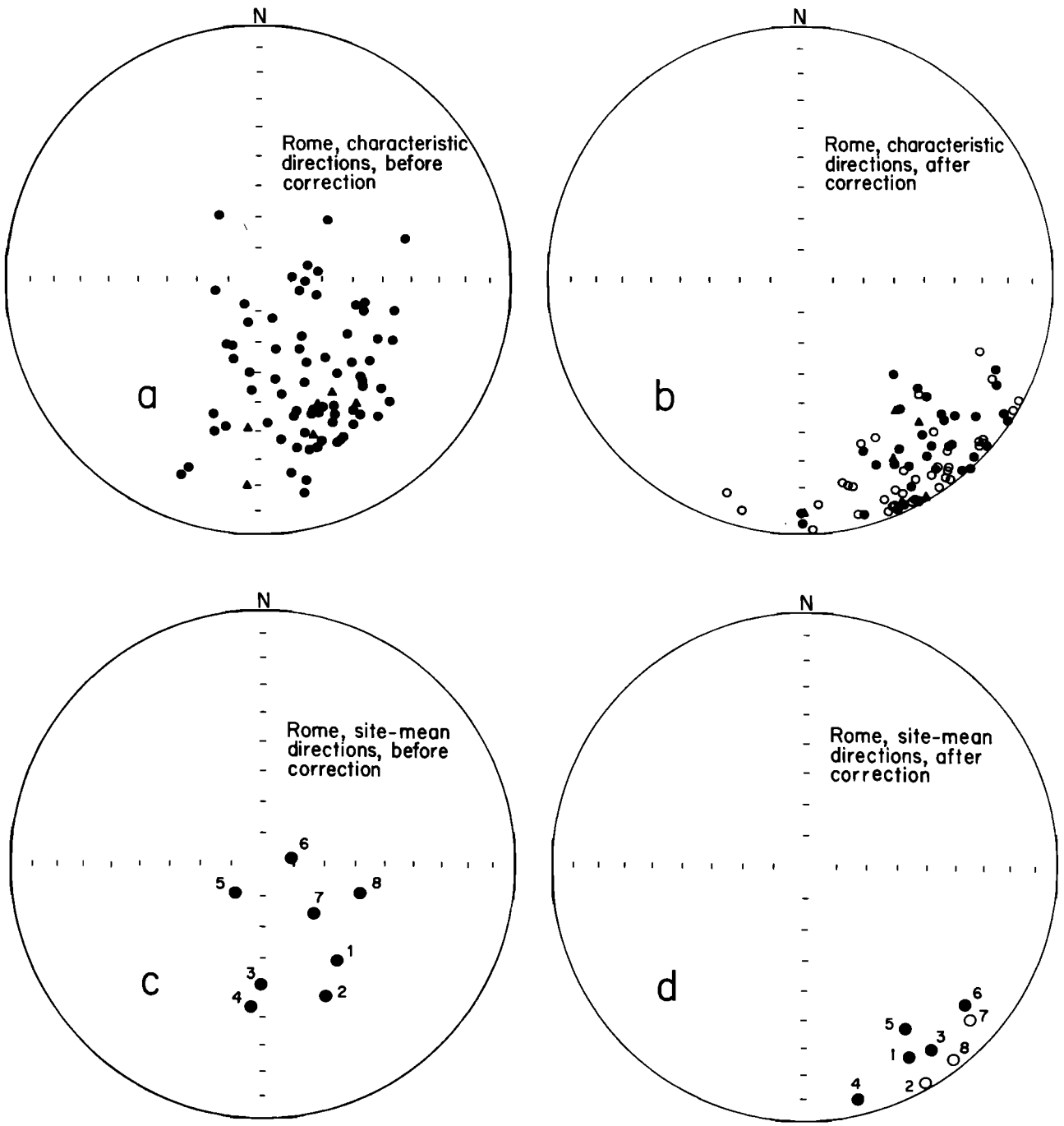


Fig. 6. Equal area projections (as in Figure 5) of the Rome Formation characteristic and site mean directions before (Figure 6a and 6c) and after (Figures 6b and 6d) correction.

No penetrative fabric except bedding is evident in the thin-sections studied. Quartz grains show various degrees of strain (undulose extinction), but the intensity of straining is highly variable and may indicate a metamorphic source terrain rather than strain of the rock.

Rotations of the Rome and Waynesboro Formations?

The Rome and Waynesboro formations are found in allochthonous structural units. Correcting for the tilt of the strata, i.e., rotation of paleomagnetic vectors about strike, gives values of pretectonic inclinations. If during orogenic deformation there were substantial rotations of parts of the Valley and Ridge Province about a vertical axis, paleomagnetic declinations would not be representative of the North American craton. The possibility of a rotation seems to be very real for the Rome Formation. A typical declination expected for our area by extrapolation from, for instance, the Early Cambrian Tapeats pole [Elston and Bressler, 1977], is to the east-southeast, whereas we observed a mean

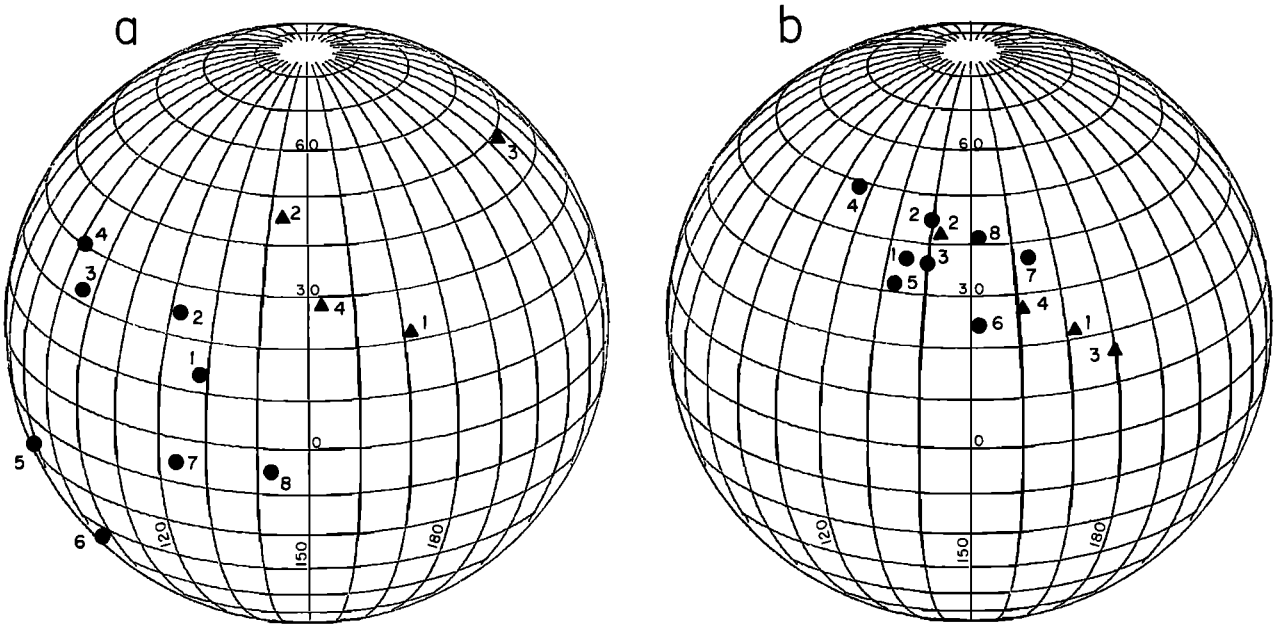


Fig. 7. Pole positions derived from site mean characteristic directions of the Rome (circles) and Waynesboro (triangles) formations. The poles before structural correction are plotted in Figure 7a and after correction in Figure 7b.

southeast declination. This would a priori suggest that if the magnetizations are coeval, the thrust sheets rotated in a clockwise sense.

Englund [1968, 1971] has studied the movement of the Pine Mountain thrust sheet (see Figure 1) relative to the Cumberland Plateau by comparing isopachs of the surrounding autochthonous plateau area. He found that the northeast end of the Pine Mountain block moved approximately 4 mi (6.4 km) with respect to the Cumberland Plateau along the Russel Fork Fault, and the southwest part of the block moved 11 mi (17.6 km) along the Jacksboro Fault. The trace of the Pine Mountain Fault is 125 mi (192 km) in length. Simple trigonometry shows therefore that a block rotation of the Pine Mountain thrust sheet could amount to about 3° with respect to the Cumberland Plateau on the basis of Englund's data. Unfortunately, this type of analysis has not been applied to other thrust sheets in the Valley and Ridge Province of Tennessee.

Evidence against relative (local) rotations between individual thrust sheets in the Valley and Ridge Province of Tennessee can be found in Ordovician paleomagnetic directions obtained by Watts and Van der Voo [1979] and in late Precambrian-Early Cambrian directions [Brown and Van der Voo, 1980]. The pretectonic paleomagnetic declinations of the contemporaneous Moccasin and Bays formations (Ordovician) agree very well, even though the units were sampled from different thrust sheets (Copper Creek and Dumplin Valley). A more regional, overall clockwise rotation of the thrust sheet sequence is also highly unlikely on the basis of the directions of Watts and Van der Voo [1979]. These directions, when compared to those of Ordovician formations of the craton (Trenton Limestone, New York State, or Beemerville Complex, New Jersey), do not deviate in a clockwise sense. If any rotations are demonstrated by these data, they would be counterclockwise, which is in a sense opposite to that required by the Rome-Tapeats comparison. Late Precambrian-Early Cambrian basalt flows of the Unicoi Formation have been sampled in northeastern Tennessee, near the North Carolina border. Characteristic directions from these basalts have westerly declinations, and their pole positions agree with contemporaneous poles [Brown and Van der Voo, 1980]. Again no rotations are demonstrated.

The site mean paleomagnetic declinations (after correction for tilt by rotation about strike) of the Rome Formation are plotted in Figure 1. The declinations are in general agreement throughout the Valley and Ridge Province of southern Virginia and Tennessee even though the data were gathered from four separate thrust sheets. Given the pretectonic age of magnetization of the Rome and the general agreement of the paleomagnetic declinations, the evidence argues against substantial relative rotations between the individual thrust sheets during or after the Alleghenian Orogeny, which was mainly responsible for the thrusting.

It has been discussed above that the Pine Mountain thrust sheet rotated, at most, 3° with respect to the North American craton. Because paleomagnetic data from other thrust sheets in Tennessee and Virginia do not favor substantial relative rotations between the blocks, we conclude that the pretectonic paleomagnetic declinations derived from this province are representative (to within a few degrees) of the paleomagnetic field of the North American craton.

Relative rotations of the Waynesboro Formation with respect to the North American craton cannot be precluded. Because of the different style of deformation the above argument cannot be applied. The site mean (corrected) paleomagnetic declinations are plotted in Figure 2. The mean declination of the Waynesboro Formation (Table 1) is

displaced eastward from the mean declination of the Rome Formation. Because of the scatter of the Waynesboro data and the possibility of age variation along strike the significance of this displacement is uncertain. Since the sampled Waynesboro strata are mostly subhorizontal, plunging fold axes alone cannot cause a deviation in declination. Van der Voo and Channell [1980] have shown that the pretectonic paleomagnetic declinations from the Upper Ordovician Juniata Formation do not show substantial clockwise rotations of the Valley and Ridge Province of Maryland, West Virginia, and Virginia, when compared with the same province in Pennsylvania, where the structural trends swing to the east. Similar conclusions were reached earlier by Roy et al. [1967] and Knowles and Opdyke [1968].

Discussion: The Rome-Waynesboro Paleomagnetic Pole

The age of magnetization characteristic of the Rome and Waynesboro formations is bracketed between the time of deposition of the units and the time of folding and thrusting of the central and southern Appalachians, i.e., roughly between the Carboniferous and the Early Cambrian. As discussed previously, there is no compelling evidence to indicate that the Rome magnetic directions have been affected by large-scale rotations about vertical axes with respect to the North American craton. It is assumed, but not conclusively demonstrated, that the Waynesboro magnetic directions, similarly, have not been affected by such rotations. The age of the characteristic magnetization is the critical factor in the interpretation of these results.

Thin-section analysis shows that hematite grains in these rocks are detrital, whereas some of the hematite fills grain interstices and coats detrital grains. It cannot be a priori assumed that the characteristic magnetization is entirely of depositional origin.

Figure 8 shows the Rome, Waynesboro, and Rome-Waynesboro mean paleomagnetic poles plotted with latest Precambrian and other Lower to Middle Cambrian poles from North America taken from Watts et al. [1980], Table 2. The poles form a streak between the extremes of the latest Precambrian poles as well as the Lower Cambrian Tapeats pole and the Middle to Upper Cambrian Muav pole. The Rome-Waynesboro pole falls in the middle of the trend. In terms of paleomagnetic directions in North America the trend is manifested by a rotation of paleomagnetic declinations with little change in inclination.

A number of explanations of this distribution are possible: (1) Several of the poles represent widespread and complete Ordovician through late Paleozoic remagnetizations of the investigated units, (2) several poles represent widespread partial (to complete) late Paleozoic remagnetizations of the sampled units (this is the remagnetization hypothesis of Creer [1968], (3) the distribution of poles may be explained by tectonic rotations of various magnitude, (4) the distribution of Cambrian poles is due to a type of geomagnetic excursion or to nondipole behavior, peculiar to the Cambrian, resulting in large swings (up to 90°) of magnetic declination

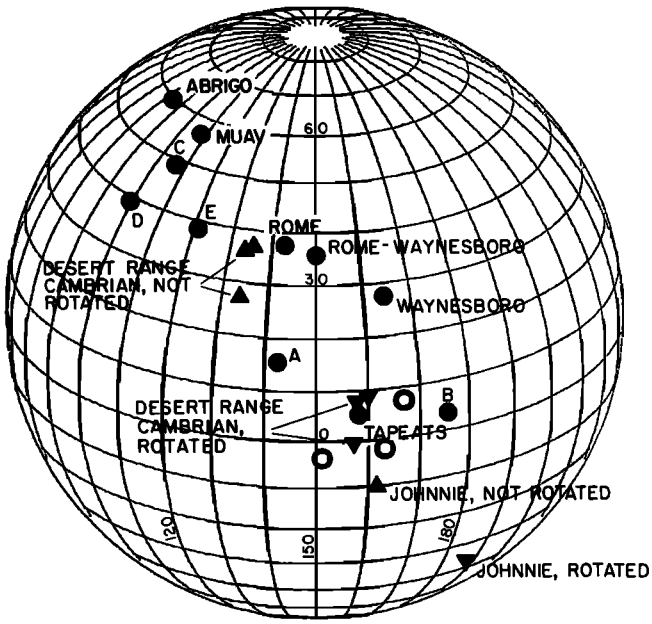


Fig. 8. Latest Precambrian (open circles) and Early to Middle Cambrian paleomagnetic poles from the North American craton [Watts et al., 1980, Table 2] and from the Desert Range of Nevada [Van Alstine and Gillett, 1979] plotted with the Rome, Waynesboro, and Rome-Waynesboro poles (Table 1). The poles from the Desert Range are triangles. The upward vertex denotes poles from the Desert Range not corrected for rotation and the downward vertex denotes poles corrected for rotation. The lettered poles are from the Colorado intrusives. Poles A through C are from French et al. [1977]. Pole D is the well-dated 525 m.y. result from Patterson et al. [1978], and pole E is from Larson and Mutschler [1971].

and little change in inclination, (5) the streak of poles characterizes the Cambrian polar wander track with respect to North America manifested as a shift in the apparent polar wander path from latest Precambrian to Middle Cambrian times, and (6) any combination of the above.

For sedimentary rocks, remagnetization (explanations 1, 2) can seldom be entirely discounted unless various attitude and petrologic tests are available [e.g., Elston and Purucker, 1979]. The Muav and Tapeats poles [Elston and Bressler, 1977] are key results in these considerations. The Muav was sampled a few meters below a Devonian unconformity, and remagnetization associated with Devonian (?) near-surface processes cannot be discounted [Elston and Bressler, 1977]. However, some of the intrusive rocks from Colorado yielded poles near to that of the Muav. The Colorado intrusives investigated by Patterson et al. [1978] have been described as unaltered and give radiometric dates of Middle Cambrian age (525 m.y.) according to the time scale of Van Eysinga [1975]. Recent Rb-Sr radiometric dating [Armbrustmacher et al., 1979] of the completely fresh alkaline complexes in Colorado studied paleomagnetically by French et al. [1977] give Middle to Late Cambrian ages of 530 to 511

m.y. for nepheline syenites and syenites. It is difficult to understand how these igneous rocks could have been remagnetized without extensive alteration of the mineralogy. Situated below the Muav limestone, the Tapeats sandstone is of late Early Cambrian to perhaps early Middle Cambrian age, and it is therefore roughly contemporaneous with the Rome Formation as well as possibly with the above-mentioned igneous rocks. The Tapeats pole was interpreted by Elston and Bressler [1977] as representative of the Early/Middle Cambrian on the basis of the magnetic stability, the occurrence of reversals, and not in the least because younger poles on the then-known North American polar wander path were largely different. However, a Late Cambrian pole from the Wilberns Formation [Van der Voo et al., 1976] falls within a degree of the Tapeats pole, and an alternative interpretation of the Tapeats pole is that it represents a remagnetization during Late Cambrian time. We will return to this point later in this section. In the case of the Rome and Waynesboro formations, remagnetization, if it occurred, would have to have been complete, as there is no evidence of a high temperature component other than that characteristic of these units. It is possible, but regarded as unlikely, that the characteristic component is the result of the simultaneous removal of two components with similar blocking temperature and coercivity spectra.

Internal tectonic rotations (explanation 3) of the platform deposits of the Grand Canyon may be rejected a priori; a major tectonic rotation of the Colorado Plateau as a whole with respect to the rest of the craton is equally unlikely and, moreover, would not resolve our problem. Tectonic rotations cannot be dismissed as an explanation for the Cambrian paleomagnetic poles from the Desert Range of Nevada obtained by Gillett and Van Alstine [1979]. They demonstrated that little relative shift of the pole occurs between the Lower Cambrian Wood Canyon Formation and the Middle Cambrian Bonanza King Formation. The Wood Canyon-Bonanza King sequence is partially coeval with the Tapeats Sandstone of the Grand Canyon. Gillett and Van Alstine [1979] assumed the magnetization of the Tapeats to be coeval with that of the Wood Canyon-Bonanza King sequence and suggested therefore a tectonic rotation of the Desert Range of Nevada as an elegant explanation for the differences between the poles. Additional geologic evidence for such rotations can be found in the local structural geology quoted by Gillett and Van Alstine [1979].

We note, however, that an apparent polar shift, similar in location and in magnitude to the (uncorrected) shift of the Nevada Sequence, is found to occur between the cratonic late Precambrian poles and the Rome/Waynesboro pole or some of the Colorado intrusive poles (Figure 8). This would render the tectonic rotation of the Nevadan Desert Range unnecessary but would raise again questions about the stratigraphic correlations or the magnetization age of the Tapeats.

Geomagnetic field excursions (explanation 4) can be discounted. A normal polarity component with northwest declination is apparently present in the Waynesboro Formation as well as in the igneous rocks from Colorado (poles A, B, and C of Figure 8 are all based on normal and reversed polarities). Furthermore, predominantly normal polarity components have recently been reported in Early to Middle Cambrian units from Labrador and the Grenville Province in Canada with pole positions very similar to poles of Figure 8. The Labrador rocks compose the late Early Cambrian Forteau Formation (G. W. Pearce, personal communication, 1979), whereas the rocks of the Grenville Province are intrusives dated as about 570-550 m.y. old [Lapointe, 1979; Dankers and Lapointe, 1979]. The concepts of normal and reversed polarities have meaning only in the context of a dipole field.

Rapid polar wander with respect to the North American craton (explanation 5) should give a systematic shift of the pole as a function of the age of the magnetizations involved. Since, however, in most instances it is the age of the rocks (stratigraphic or radiometric) rather than the age of the magnetization acquisition that is known, we must infer these to be the same. The Rome Formation is of latest Early Cambrian age [Rodgers, 1953; Milici, 1973] and the Waynesboro Formation is regarded to be of similar age [King, 1950; Cloos, 1971]. The Tapeats is of late Early Cambrian or early Middle Cambrian age. The Wood Canyon, Carrara, and Bonanza King formations of the Desert Range of Nevada apparently span this same interval of time. It is well documented that little apparent polar wander occurred between the times the latter units acquired their characteristic directions. If apparent polar wander during the Cambrian did in fact occur, then the magnetization ages of the above units are called into question.

In summary, it is clear that none of the five explanations can do justice to all lower and Middle Cambrian poles, with the currently known age relationships. Ad hoc explanations invoking rotations, remagnetizations, or nondipole behavior can be found for subsets of some Cambrian poles but conflict with other subsets. We conclude that at the present level of analysis the problems with the observed pattern of North American Cambrian paleomagnetic poles can be reduced to a few plausible explanations, and there are difficulties with each. However, because the pattern is recorded in rocks of different lithologies and histories, from diverse parts of the North American craton, it is suggested that the simplest, most unifying hypothesis explaining this pattern is that of apparent polar wander with respect to the North American craton during the Cambrian.

For the time span of the latest Precambrian and the Lower and Middle Cambrian this apparent polar wander entails a gradual shift of the pole from the equator at about 160°E to about 45°N, 110°E or beyond. In a companion paper we will address the late Middle and late Cambrian poles and argue that a complete Cambrian loop is necessary because of the Late Cambrian poles from the Llano uplift.

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References

Amato, R. V., Geology of the Salem Quadrangle, Virginia, Va. Div. Miner. Resour. Rep. Invest., 37, 40 pp., 1974.

Armbrustmacher, T. J., C. E. Hedge, and R. L. Parker, Alkaline rock complexes in the Wet Mountains area, Fremont and Custer counties, Colorado: Genetic implications of minor-element and Sr-isotope geochemistry, Geol. Soc. Amer. Abstr. Program, 11, 380-381, 1979.

Brown, P., and R. Van der Voo, Paleomagnetism of the Cambrian Unicoi basalts, Blue Ridge, Tennessee, and Virginia, Eos Trans. AGU, 61, 219, 1980.

Cloos, E., The geology of the South Mountain anticlinorium, Maryland, Guidebook Vol. 1, Stud. in Geol., 16, part 1, pp. 1-28, Johns Hopkins Univ., Baltimore, Md., 1950.

Cloos, E., Structural geology of South Mountain and Appalachians in Maryland, Guidebook, Vols. 4 and 5, Stud. Geol.. 17, pp. 1-76, Johns Hopkins Univ., Baltimore, Md., 1958.

Cloos, E., Microtectonics along the Western Edge of the Blue Ridge, Maryland and Virginia, 234 pp., Johns Hopkins Press, Baltimore, Md., 1971.

Creer, K. M., Paleozoic paleomagnetism, Nature, 219, 246-250, 1968.

Dankers, P., and P. L. Lapointe, Paleomagnetism of Grenville diabases and volcanics (abstract), Eos Trans. AGU, 60, 749, 1979.

Elston, D. P., and S. L. Bressler, Paleomagnetic poles and polarity zonation from Cambrian and Devonian strata of Arizona, Earth Planet. Sci. Lett., 36, 423-433, 1977.

Elston, D. P., and M. E. Purucker, Detrital magnetization in red beds of the Moenkopi Formation (Triassic), Gray Mountain, Arizona, J. Geophys. Res., 84, 1653-1666, 1979.

Englund, K. J., Geology and coal resources of the Elk Valley Area, Tennessee and Kentucky, U.S. Geol. Surv. Prof. Pap., 572, 59 pp., 1968.

Englund, K. J., Displacement of the Pocahontas Formation by the Russell Fork Fault, Southwest Virginia, U.S. Geol. Surv. Prof. Pap., 750-B, B13-B16, 1971.

French, R. B., Lower Paleozoic paleomagnetism of the North American craton, Ph.D. thesis, 150 pp., Univ. of Michigan, Ann Arbor, 1976.

French, R. B., D. H. Alexander, and R. Van der Voo, Paleomagnetism of upper Precambrian to lower Paleozoic intrusive rocks from Colorado, Geol. Soc. Amer. Bull., 88, 1785-1792, 1977.

Gathright, T. M., and P. G. Nystrom, Geology of the Ashby Gap Quadrangle, Virginia, Va. Div. Miner. Resour. Rep. Invest., 36, 55 pp., 1974.

Gillett, S. L., and D. R. Van Alstine, Paleomagnetism of Lower and Middle Cambrian sedimentary rocks from the Desert Range, Nevada, J. Geophys. Res., 84, 4475-4489, 1979.

Henry. S. G., Chemical demagnetization: Method, procedures and applications through vector analysis, Can. J. Earth Sci., 16, 1832-1841, 1979.

King, P. B., Geology of the Elkton area, Virginia, U.S. Geol. Surv. Prof. Pap., 230, 82 pp., 1950.

Knowles, R. R., and N. D. Opdyke, Paleomagnetic results from the Mauch Chunk Formation: A test of the origin of curvature in the folded Appalachians of Pennsylvania, J. Geophys. Res., 73, 6515-6526, 1968.

Lapointe, P. L., Paleomagnetism of the Sept-Iles anorthosite and the tectonic history of the Appalachians (abstract), Eos Trans. AGU, 60, 749, 1979.

Larson, E. E., and F. E. Mutschler, Anomalous paleomagnetic pole from isotopically dated Cambro-Ordovician intrusives in Colorado, Geol. Soc. Amer. Bull., 82, 1657-1666, 1971.

McElhinny, M. W., Statistical significance of the fold test in Paleomagnetism, Geophys. J. Roy. Astron. Soc., 8, 338-340, 1964.

McElhinny, M. W., Paleomagnetism and Plate Tectonics, 358 pp., Cambridge University Press, New York, 1973.

McGuire, O. S., Geology of the Daleville Quadrangle, Virginia, Va. Div. Miner. Resour. Rep. Invest., 42, 43 pp., 1976.

Milici, R. C., The stratigraphy of Knox County, Tennessee, Geology of Knox County, Bull. 70, pp. 9-24, Tenn. Div. of Geol., Nashville, 1973.

Patterson, P. E., J. G. Rosenbaum, and E. E. Larson, Refinement of a Cambro-Ordovician paleomagnetic pole from plutonic rocks in southwestern Colorado, Eos Trans. AGU, 59, 1060, 1978.

Rodgers, J., Geologic map of east Tennessee with explanatory text, Bull. 58, 168 pp., Tenn. Div. of Geol., Nashville, 1953.

Roy, J. L., N. D. Opdyke, and E. Irving, Further paleomagnetic results from the Bloomsburg Formation, J. Geophys. Res., 72, 5075-5086. 1967.

Slaughter, T. H., and J. M. Darling, The water resources of Allegany and Washington counties, Bull. 24, 408 pp., Dept. of Geol. Mines and Water Resour., Baltimore, Md., 1962.

Stose, G. W., The sedimentary rocks of South Mountain, Pennsylvania, J. Geol. 14, 201-220, 1906.

Swingle, G. D., Structural geology of Knox County, Tennessee, Geology of Knox County, Bull. 70, pp. 63-73, Tenn. Div. of Geol., Nashville, 1973.

Van Alstine, D. R., and S. L. Gillett, Paleomagnetism of Upper Precambrian sedimentary rocks from the Desert Range, Nevada, J. Geophys. Res. 84, 4490-4500, 1979.

Van der Voo, R., The Cambrian paleomagnetic field for the North American craton, Eos Trans. AGU, 59, 263-264, 1978.

Van der Voo, R., and J. E. T. Channell, Paleomagnetism in orogenic belts, Rev. Geophys. Space Phys., 18, 455-481, 1980.

Van der Voo, R., and R. B. French, Paleomagnetism of the Late Ordovician Juniata Formation and the remagnetization hypothesis, J. Geophys. Res., 82, 5796-5802, 1977.

Van der Voo, R., R. B. French, and D. W. Williams, Paleomagnetism of the Wilberns Formation (Texas), and the Late Cambrian paleomagnetic field for North America, J. Geophys. Res., 81, 5633-5638, 1976.

Van Eysinga, F. W. B., Geological Time Table, Elsevier, New York, 1975.

Watts, D. R., and R. Van der Voo, Paleomagnetic results from the Ordovician Moccasin, Bays, and Chapman Ridge formations of the Valley and Ridge Province, eastern Tennessee, J. Geophys. Res., 84, 645-655, 1979.

Watts, D. R., Van der Voo, and S. C. Reeve, Cambrian paleomagnetism of the Llano uplift, Texas, J. Geophys. Res., 85, this issue, 1980.
Zijderveld, J. D. A., A. C. demagnetization of rocks: Analysis of results, in Methods of Paleomagnetism, edited by D. W. Collinson, K. M. Creer, and S. K. Runcorn, pp. 254-286, Elsevier, New York, 1967.

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